

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-10: Closest Pair of Points – Brute Force

Aim:

To find the pair of points having the minimum distance between them using the brute-force technique.

Algorithm:

1. Read the number of points.
2. Read the coordinates (x, y) of each point.
3. Compare every pair of points.
4. Calculate the Euclidean distance between each pair.
5. Keep track of the pair having the minimum distance.
6. Display the closest pair and its distance.
7. Display the time and space complexity.

Source Code:

```
import math
```

```
n = int(input("Enter number of points: "))
```

```
points = []
```

```
for i in range(n):
```

```
    x, y = map(float, input("Enter x and y coordinates: ").split())
```

```
    points.append((x, y))
```

```
min_distance = float('inf')
closest_pair = None
for i in range(n):
    for j in range(i + 1, n):
        x1, y1 = points[i]
        x2, y2 = points[j]
        distance = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)
        if distance < min_distance:
            min_distance = distance
            closest_pair = (points[i], points[j])
print("Closest Pair:", closest_pair)
print("Minimum Distance:", round(min_distance, 2))
print("Time Complexity: O(n^2)")
print("Space Complexity: O(n)")
```

Output:

```
Enter number of points: 4
Enter x and y coordinates: 0 0
Enter x and y coordinates: 5 5
Enter x and y coordinates: 1 1
Enter x and y coordinates: 10 10
Closest Pair: ((0.0, 0.0), (1.0, 1.0))
Minimum Distance: 1.41
Time Complexity: O(n^2)
Space Complexity: O(n)
```

Result:

The closest pair of points was successfully found using the brute-force approach.